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SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Jasmitha R (192521295) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Jasmitha.R

17/20

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem.
   Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Introduction:

A smart home automation system controls lighting temperature, and security automatically. It uses sensors such as motion sensors, temperature sensors, and clocks to make decisions based on the environment and user preference.

Types of Intelligence Agents:-

1. Simple Reflex Agent:-
   - Acts only on the current input.
   - Uses condition - action rules
   - Does not remember previous events.

Eg:-

- If motion is detected → TURN ON the light
- If no motion is detected → Turn OFF the lig

Advantage:

- Fast response
- Easy to implement

Disadvantage:

- Cannot learn user habits
- cannot handle complex situations.

2. Model-Based Reflex Agent:-

- Maintains an Internal model of the environment.
- Uses current and previous einformation.

Eg:-

- Remembers that the bedroom light is already ON
- keeps the room temperature between 22-24°C

Advantages:-

- Better decision making
- works even with incomplete information.

Disadvantages:-

- More memory required
- More complex than simple reflex agents.

3. Goal-Based Agent:-

- Makes decision to achieve a specific goal.
- plans actions before executing them.

Eg:-

- keep the house secure
- Reduce electricity consumption
- Maintain comfortable temperature

Advantages:-

- Intelligent planning
- Flexible decision making

A Robot must find the exit of an unknown maze without knowing its layout. Since there is no prior knowledge, Uninformed search Algorithms such as uniform cost search (ucs) and Iterative Deepening search (IDS) are used.

1. Uniform cost search (UCS)

working:-

- Expands the node with the lowest path cost first.
- Uses a priority queue.
- continues until the goal is reached.

Advantages:-

- Finds the minimum-cost path
- complete and optimal

Disadvantages:-

- High memory usage.
- slow for large search spaces

complexity.

- Time:-

$O(b^{1} + [C^{*}/E]$

Disadvantages:-

- Requires more computation
- Slower than reflex agent

4. Utility - Based Agent.

- choose the action that provides the highest benefit.
- considers multiple factors such as comfort cost, and energy.

Eg:- Instead of keeping the AC at 18°C, it sets it to 24°c because it saves electricity while keeping the room comfortable.

Advantages:-

- Best overall decision
- Balances comfort - security, and energy savings.

Disadvantages

- utility calculation is complex
- Requires more processing power.

conclusion:-

It is analyzed how the each type d function within the system ensures ort - energy efficiency and security.

- Flexible decision making

Disadvantages:-

- Requires more computation
- Slower than reflex agent

4. Utility - Based Agent.

- choose the action that provides the highest benefit.
- Considers multiple factors such as comfort cost, and energy.

Eg:- Instead of keeping the AC at 18°C, it sets it to 24°c because it saves electricity while keeping the room comfortable.

Advantages:-

- Best overall decision
- Balances comfort, security, and energy savings.

Disadvantages

- utility calculation is complex
- Requires more processing power.

Conclusion:-

It is analyzed how the each type would function within the system ensures comfort, energy efficiency and security.

2. Iterative Deepening :-

* It performs through the Iterative algorithm.

Advantanges:

* It is fast access
* It is reliable and store the information
* It is programmable.

Disadvantages:-

* Calculation will be complex
* The stored data will be accessable.
* This will be not easy access.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with majo gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written difficult unders d